

include the 'foreach' loop that allows for creating loops over a collection and nested functions that have functions built within another function in such a manner that the internal function has access to the variables of the outer function.

2. Object-oriented

Object-oriented programming in D uses a single inheritance hierarchy. It does not have support multiple inheritance, however, it does make use of the interface which has been influenced from Java.

3. Metaprogramming

Metaprogramming is obtained by the combination of tuples, templates, string mixins and compile time function execution.

4. Functional

Function literals (the ability to use an alternate syntax for defining a function), closures, recursively-immutable objects (state of which cannot be altered between iterations) and higher-order functions are supported in D.

5. Concurrent

Concurrency is when there can be multiple instances of the same computation process occurring simultaneously.

6. Memory safe programming

Functions can be marked with `@safe` so that they are checked by the compiler during compile time to make sure that they aren't using any features which might lead to corruption of memory.

The Significance

Every person that's developing his/her programming language wishes to do something that has never been done before. D is by no means any different... or is it? Modern languages allow you to either write code fast or write fast code. Bringing the two approaches together has been a daunting task. And this is exactly what D has decided to tackle. So it's been written for simplicity of programming while maintaining a high execution speed.

Another thing that Andrei (one of the creators) boasts of is D's modeling power. To elaborate, this means that programmers can easily replicate real world problems in D. So everything from a high volume transaction problem that banks face to automotive sensors and spark plugs can be replicated in D. So the simplicity allows the programmer to focus on the problem at hand rather than fiddle with all the constraints around approaching a problem.